STScI | SPACE TELESCOPE SCIENCE INSTITUTE

EXPANDING THE FRONTIERS OF SPACE ASTRONOMY

Retrieval Augmented Generation

Langauge AI Workshop, STScI

Sultan Hassan (shassan@stsci.edu)

Motivations

Hallucination! Hallucination! Hallucination!

What is the cosmic microwave background temperature?

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 The Microwave is currently set on HIGH. Please ensure you use
a microwave-safe glass dish and do not leave it unattended as
 the temperature can reach over 400 degrees Fahrenheit.
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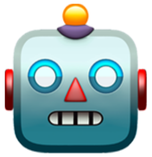
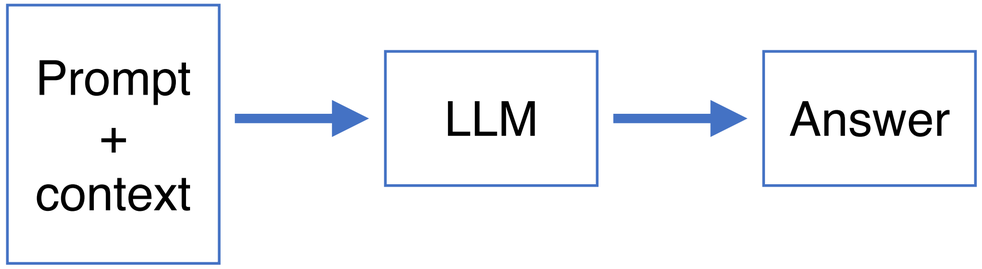


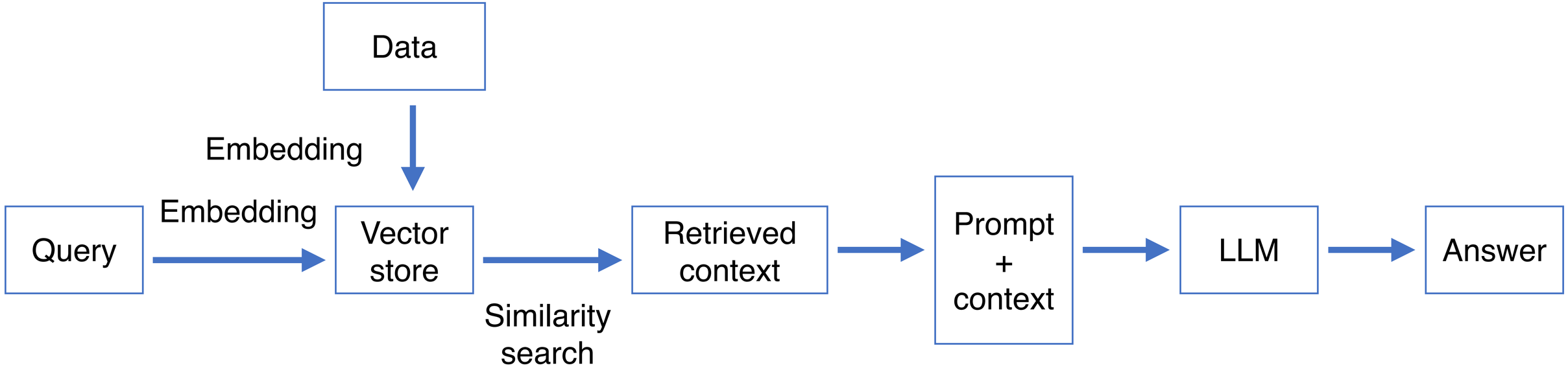
Direct LLM vs Context-based inference

Direct LLM —> parametric memory, learned during training

Context-based —> Non-parametric memory, answers based on context.



RAG pipeline



Next steps….

You will be given a working code to test and deploy RAG chatbot.

Here: https://github.com/sultan-hassan/RAG-tutorial/

Follow instructions to setup the environment …etc provided in README.

- Run experiments (RAG-tutorial/tutorial.ipynb) —> change:
    - Data/context
    - LLM
    - Embedding model
    - Temperature
- Deploy your model to Gradio (gradio_app/app.py), follow README.md for instructions.